OC-GS: Object-Centric Representation–Demand Alignment for Immersive 3D Gaussian Streaming

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Abstract

3D Gaussian Splatting (3DGS) has emerged as the definitive representation for next-generation immersive media, yet its practical delivery over resource-constrained networks remains fundamentally unresolved. We argue that this is not merely a bandwidth or compression challenge, but a **representation–demand alignment** problem: existing systems organize transmission around spatial proximity, while user attention in immersive environments is organized around semantic objects—producing a structural mismatch at the representation, prediction, and control levels that cannot be resolved by bandwidth alone.

To correct this alignment failure, we present **OC-GS** (Object-Centric Gaussian Streaming), a two-stage alignment framework built around an *object-addressable streaming substrate*. **Stage 1 (Representation Realignment):** A Semantic-Saliency Co-distillation pipeline transforms raw Gaussians into an object-addressable substrate, elevating semantics from a passive scene attribute to an active *semantic utility* signal for resource allocation. A Hybrid Granularity architecture preserves spatial octrees for the macro-environment while packaging high-value entities as atomic micro-object bundles, fully decoupling semantic targets from their geometric context. **Stage 2 (Stability-Governed Delivery):** An object-grounded probabilistic predictor fuses physical kinematics, visual geometry, and a fixation-token intention prior to produce an object-level demand distribution, which feeds a Lyapunov-based solver that provides provable stability guarantees under coupled bandwidth and buffer constraints—achieving *stability-governed* streaming with churn suppression by design. Extensive evaluations on eye-tracked 6DoF traces demonstrate that OC-GS reduces redundant transmission by [XX]% and improves Foveated-PSNR by [XX] dB compared to state-of-the-art spatial-adaptive baselines, while achieving near-zero refetch ratios under realistic 5G network conditions.

Keywords

3D Gaussian Splatting, Representation–Demand Alignment, Object-Addressable Streaming Substrate, Semantic Utility, Stability-Governed Delivery, 6DoF Immersive Media

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1 Introduction

2 Introduction

With its photorealistic appearance and real-time rendering capability, 3D Gaussian Splatting (3DGS) [12] has become a compelling representation for next-generation immersive media. More broadly, it inherits a line of radiance-field systems that progressively improved view-synthesis quality, explicitness, and rendering efficiency, including NeRF, PlenOctrees, Plenoxels, and Instant-NGP [9, 19, 21, 42]. Its native support for 6-Degree-of-Freedom (6DoF) interaction is particularly attractive for VR/AR applications, where users inspect scenes from continuously changing viewpoints. Yet streaming high-fidelity 3DGS over real networks remains difficult: a single scene may contain gigabytes of Gaussian parameters, the visible set shifts rapidly under head motion, and eye movements further tighten the latency budget for perceptually important content. Together, these factors place simultaneous pressure on bandwidth, client memory, and rendering resources.

Under this setting, one design choice becomes central: *once the system knows where the user is looking, what should be the atomic unit of transmission, caching, and quality allocation within the FoV?* This question is crucial because immersive QoE depends not only on excluding out-of-view content, but also on whether limited runtime resources can be concentrated on the small set of entities that dominate perception at each moment.

Existing 3DGS streaming systems answer this question largely in *spatial* terms. They use FoV or view-dependent visibility for coarse pruning, and then progressively stream spatial partitions such as tiles, grids, or hierarchical nodes [10, 29, 31, 36, 43]. This design scales well and simplifies geometric indexing, but it implicitly assumes that spatial proximity is a reasonable proxy for perceptual demand.

Our analysis of eye-tracked 6DoF traces (§4.1) shows that this assumption often fails in immersive viewing. User attention is strongly **object-centric**: viewers repeatedly inspect semantic entities rather than arbitrary spatial coordinates. This creates a fundamental **representation–demand mismatch**: the streaming substrate is organized by space, whereas perceptual demand is organized by objects. Once gaze is introduced for fine-grained prioritization inside the FoV, this mismatch becomes a first-order limitation rather than a minor inefficiency.

This mismatch manifests in three closely related challenges. **(C1: Granularity mismatch.)** Spatial containers entangle a salient object with surrounding clutter, so refining the target still incurs substantial contextual overhead and may break object-level LoD coherence. **(C2: Prediction lag.)** Because scheduling remains largely reactive to the current FoV or gaze sample, the system often trails

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fast object-directed attention and struggles to satisfy the short post-saccade comfort window. **(C3: Control instability.)** Under coupled bandwidth, cache, and rendering constraints, even small errors in predicted demand can trigger repeated load–evict–refetch cycles, leading to cache churn and visible quality oscillation.

To address these issues, we present **OC-GS**, a two-stage object-centric streaming framework. In the offline stage, a **Semantic-Saliency Co-distillation** pipeline transforms a frozen 3DGS scene into an *object-addressable* asset, while a **Hybrid Granularity** representation preserves background geometry in macro spatial octrees and promotes important entities to independently streamable micro-object bundles. In the online stage, a **Neuro-Symbolic Controller** predicts object-level demand from motion, visibility, and intention cues, and a **Lyapunov-based solver** converts these predictions into stable fetching and refinement decisions. OC-GS therefore retains FoV-based pruning for coarse visibility reduction, but performs fine-grained transmission control in object space, bringing the delivery pipeline into closer alignment with how users actually allocate attention.

Our contributions are three-fold:

- **An empirical and cognitive foundation for object-centric demand.** Through a trace-driven analysis of user viewing preferences, corroborated by findings from object-based attention, cognitive psychology, and neuropsychological studies, we show that user attention in immersive environments is concentrated primarily on semantic objects rather than abstract spatial coordinates. This establishes the empirical basis for the object-centric nature of immersive demand and motivates our representation–demand alignment perspective.
- **An object-addressable streaming substrate.** We introduce a Semantic-Saliency Co-distillation pipeline that elevates semantics to an active *semantic utility* signal, and a Hybrid Granularity architecture that constructs an object-addressable substrate from any frozen 3DGS scene, eliminating contextual waste by fully decoupling semantic objects from spatial partitioning.
- **A stability-governed delivery principle.** We integrate object-level probabilistic prediction with Lyapunov optimization to achieve *stability-governed* demand-aware delivery, providing provable queue stability and churn suppression under coupled bandwidth and buffer constraints.

3 Related Work

3.1 3D Gaussian Splatting and Compression

3D Gaussian Splatting (3DGS)[12] has substantially advanced real-time radiance field rendering by coupling the flexibility of point-based representations with a differentiable splatting rasterization pipeline. It also builds on a broader radiance-field trajectory spanning NeRF [19], explicit acceleration structures such as PlenOctrees [42] and Plenoxels [9], and highly optimized real-time implementations such as Instant-NGP [21]. However, its explicit parameterization yields massive storage footprints—frequently at the gigabyte scale per scene (e.g., 1.5 GB reported for a single delivery scenario[31])—posing stringent challenges for transmission. To mitigate this, recent works have focused on compression and compact representations, including parameter quantization and learned codebooks[22, 34], as well as reducing the memory footprint while retaining rendering compatibility[23]. While these methods substantially reduce storage overhead, they are primarily designed for *static* compression and assume the full representation is available prior to interactive rendering, precluding progressive, low-latency random access as required for streaming[44]. In contrast, **OC-GS** targets a *streaming-ready* representation where geometry (Base Layer) and appearance (Spectral Layer) are decoupled, allowing the client to render a structural proxy immediately while progressively fetching spectral details without global decompression blocking.

3.2 LOD Construction and Streaming for 3DGS

To enable web-based or mobile delivery, researchers have introduced Level-of-Detail (LoD) and spatial partitioning to 3DGS. Common approaches adapt hierarchical structures from large-scale rendering: Octree-GS[26] and LODGE[14] organize Gaussians into multi-resolution or chunked nodes, adjusting the active set based on viewpoint/scale, while LOD-GS explores scalable LoD generation with a soup-style representation[28, 35]. Beyond rendering-side LoD, systems for 3DGS delivery also investigate progressive/layered scene transmission and scheduling[31, 44]. However, these approaches invariably rely on *spatial discretization* as the atomic scheduling unit. As detailed in §4, such partitioning disregards object boundaries; transitions between nodes, chunks, or layers can introduce perceptible temporal discontinuities, necessitating explicit mitigation mechanisms such as opacity blending[14]. Furthermore, most simplification pipelines couple geometry and appearance at identical granularity, making it structurally difficult to stabilize geometric content before progressively refining appearance. Our work diverges by constructing an **object-addressable streaming substrate**: we treat semantic objects as the minimum scheduling unit, expose each object's *semantic utility* as an explicit resource-allocation signal, and enforce a strict separation between geometric stability and spectral refinement—directly correcting the representation–demand alignment failure that spatial-first systems inherit by design.

3.3 Gaze-Contingent and Semantic-Aware Rendering

Gaze-contingent rendering exploits the non-uniform acuity of human vision to save resources. In rasterization and real-time graphics, foveated rendering reduces shading complexity in the visual periphery[24]. In the context of neural rendering, FoV-NeRF[4] pioneered gaze-contingent radiance fields, and recent works have explored gaze-tracked foveated rendering with 3DGS and related point-based pipelines[7, 8, 36]. On the prediction side, prior immersive-media studies have modeled future gaze or head movement in panoramic content [38, 39], but these predictors are typically optimized in view or trajectory space rather than object space. However, most existing systems map gaze strictly to spatial coordinates (e.g., radial masks). They fail to account for the cognitive mechanism of **object-based attention**[6, 27], where attention tends to spread within a coherent object rather than staying confined to a fixed-radius disk. On the other hand, promptable segmentation and semantic 3DGS pipelines, including Segment Anything, SAM 2, LangSplat,

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and Gaussian Grouping[13, 15, 25, 40], embed semantics into visual representations for querying, editing, or segmentation, but they have not leveraged these semantics for bandwidth orchestration. **OC-GS** bridges this gap by reframing semantic enrichment as an alignment problem. Rather than treating semantics as passive annotations, we elevate them into active *semantic utility* signals that drive both the construction of an *object-addressable streaming substrate* at the offline stage and *stability-governed delivery* decisions at runtime.

4 Motivation

4.1 Object-Centric Attention and Temporal Repeatability

Attention is organized around coherent objects rather than isolated spatial points. A large body of object-based attention research shows that attentional selection and facilitation are often *object-based*: attention is sustained over discrete object representations and tends to be coherent within an object's boundaries, rather than acting as an independent boost at an arbitrary spatial coordinate [5, 27, 32, 41]. Classic cueing studies provide operational evidence. In the Egly–Driver–Rafal "two-rectangles" paradigm [6], targets appearing on the cued object are detected faster than equidistant targets on another object, a phenomenon known as *the same object advantage*. This effect suggests that facilitation propagates along object structure rather than being tied solely to a point in space [6, 33]. Eye-movement and transsaccadic studies further show that observers preferentially continue gaze within the same object and maintain object continuity across saccades [11, 30]. This object-centric view is also echoed in immersive media systems, such as object-based broadcasting, which advocate structuring content as manipulable objects assembled at the receiver. [1, 2]. *Consequently, in immersive viewing, user's attention tends to concentrate on a small set of semantically meaningful entities, rather than being uniformly distributed across the FoV.*

Our eye-tracked traces reveal strong temporal repeatability of attended content. Attended regions repeatedly reappear over long horizons with only small deviations, revealing *revisit locality* that a streaming system can exploit. This temporal locality is a core observation behind cache-assisted six-degree-of-freedom streaming [16], and we observe it consistently in our setting. For each frame t, we construct a gaze-centered ROI mask A_t on the image plane, such as a foveal disk or a Gaussian-weighted neighborhood around the 2D gaze point. We then measure repeatability as the expected overlap between ROIs separated by a time lag Δ:

$$\mathrm{Rep}(\Delta) = \mathbb{E}_t\Big[\frac{|A_t \cap A_{t+\Delta}|}{|A_{t+\Delta}|}\Big] \tag{1}$$

Fig. 1(a) plots $\mathrm{Rep}(\Delta)$ across a wide range of time lags. We observe a clear non-zero plateau that persists for tens of seconds, showing that a meaningful fraction of future attended regions still overlaps with previously attended regions even at large Δ. The close agreement between left-eye and right-eye traces further indicates that this repeatability is not driven by single-eye noise. Consequently, immersive viewing is not memoryless, and high-utility content reappears repeatedly over long horizons.

4.2 Temporal Repeatability and Caching Headroom

Temporal repeatability implies substantial headroom to improve long-horizon utility via caching. To quantify this potential without committing to a specific online controller, we perform a trace-driven replay analysis following the *measure-then-exploit* methodology in cache-assisted immersive streaming [16]. We discretize each frame into coarse image-plane blocks and define the per-epoch high-value set S_t as the blocks that intersect the gaze ROI. Optionally, each attended block $b \in S_t$ is assigned a foveation-inspired weight $w_{t,b}$ that decays with its distance to the gaze center, assigning higher weight near the center [37].

The replay model assumes a fixed per-epoch download budget. At epoch t, the client may fetch a set $\mathcal{F}_t$ of enhancement blocks under this budget while reusing the blocks already stored in the cache $\mathcal{C}_t$. The blocks available for high-quality rendering at epoch t are

$$\mathcal{A}_t \;=\; \mathcal{C}_t \cup \mathcal{F}_t \tag{2}$$

which induces the per-epoch attended utility

$$u_t \;=\; \sum_{b \in S_t} w_{t,b} \cdot \mathbb{1}[b \in \mathcal{A}_t]\,, \qquad U(c) \;=\; \sum_t u_t \tag{3}$$

Here c denotes the cache capacity measured in the number of enhancement blocks that can be stored. We report relative long-horizon utility normalized by the no-cache baseline ($c = 0$):

$$\widetilde{U}(c) \;=\; \frac{U(c)}{U(0)} \tag{4}$$

so that $\widetilde{U} = 1$ corresponds to **No-cache**.

Fig. 1(b) compares **No-cache**, a simple Least Recently Used (**LRU**) cache, and an **Ideal** lookahead replacement policy that leverages future accesses in the trace to implement the offline-optimal replacement, serving as an upper bound on achievable caching gains. LRU is a recency-based baseline: when the cache is full, it evicts the block whose *most recent* access is the farthest in the past. The x-axis shows cache capacity expressed as a fraction of enhancement blocks, and the y-axis shows relative accumulated attended utility with No-cache normalized to one. The gap between LRU and Ideal quantifies the headroom: revisit locality can translate into substantially higher long-horizon utility if the system makes revisit-aware keep/evict/prefetch decisions rather than relying on short-sighted recency alone.

Consequently, temporal repeatability is not only a behavioral phenomenon but also a first-order systems opportunity. It suggests that a small amount of carefully retained content can cover a large fraction of future attended regions, while naive recency-based caching may fail under long-horizon revisits. This motivates an online scheduler that explicitly accounts for revisit locality, combining gaze-driven demand, resource constraints, and stability considerations, to decide what to fetch and what to keep.

4.3 Caching Thrashing under Multi-Timescale Gaze Dynamics

Even during fixation, gaze dynamics can trigger cache churn, preventing naive caching from realizing the headroom. The

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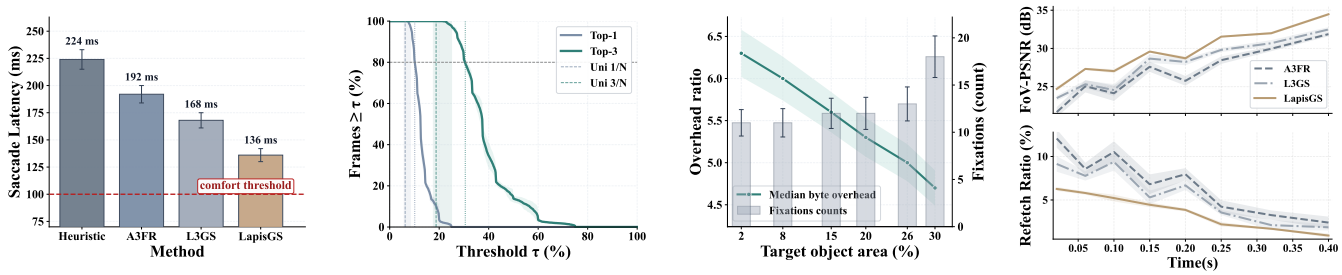

Figure 1: Figure 1: Trace-driven motivation. We analyze eye-tracked 6DoF traces to reveal why spatial, cache-centric heuristics are insufficient for interactive 3DGS streaming. (a) Object-centric attention concentration: the share of frames whose Top-1 and Top-3 objects capture at least a threshold τ of foveated attention, contrasted with a uniform baseline $\frac{K}{N_t}$ where N_t is the number of visible instances in frame t. (b) Chunk-size mismatch: spatially coupled chunks incur high byte overhead to refine a small attended target; the line shows median overhead versus target projected area, and bars report fixation counts per area bin. (c) Saccade-latency bottleneck: representative baselines (heuristic reactive fetching, A3FR, L3GS, LapisGS) fail to meet a 100 ms comfort budget for post-saccade quality recovery. (d) Instability under uncertainty: per-epoch quality (FoV-PSNR) and refetch ratio fluctuate over time under spatial-first policies, showing that coupled bandwidth/cache constraints can trigger churn and quality drops. These observations motivate an object-addressable streaming substrate with uncertainty-aware prediction and stability-aware online control.

headroom revealed by the Ideal lookahead policy does not automatically translate into gains for an online cache. A key reason is that the attended set S_t is not stationary: even during nominal fixation, gaze exhibits persistent micro-movements (e.g., microsaccades) and tracking noise [17, 18, 20], which repeatedly perturbs the ROI boundary and causes blocks near the boundary to intermittently enter and leave S_t. Under limited cache capacity and a fixed per-epoch fetch budget, a purely recency-based policy such as LRU can therefore enter a churn cycle: blocks are evicted when they briefly fall outside S_t, yet are needed again shortly afterward when the ROI jitters back.

We quantify this thrashing via trace-driven replay of cache events. Let E_t and R_t denote the numbers of evictions and refetches at epoch t, where a *refetch* means downloading a block that was previously cached but is missing when requested again. We define the refetch fraction

$$\eta_t = \frac{R_t}{E_t + \epsilon} \tag{5}$$

which measures what fraction of cache turnovers are rapidly reversed by revisits and thus directly indicates churn. Fig. 1(c) reports η_t under two subsets of epochs: *all epochs* and *fixation-only* epochs. Notably, the fixation-only curve closely tracks the all-epoch trend, indicating that churn arises not only from large saccadic displacements but persists even during sustained fixation. Consequently, a recency-based policy such as LRU spends a substantial fraction of time with high η_t, implying that a significant portion of download budget is consumed by re-acquiring recently evicted content rather than extending high-quality coverage. *Consequently, realizing the long-horizon headroom requires cache management that explicitly accounts for revisit likelihood and temporal stability, rather than relying on short-sighted recency alone.*

4.4 Utility Instability from Churn under Coupled Constraints

Churn manifests as unstable quality evolution, motivating stability-aware online control under coupled constraints. The preceding subsections have established that (i) long-horizon revisit locality creates substantial caching headroom, yet (ii) naive recency-based caching suffers from churn even during nominal fixation. We now demonstrate that such churn is not an isolated cache-level artifact: it propagates directly into *visible* and *persistent* quality instability at the rendered output.

We instantiate the per-epoch utility as rendered quality measured by PSNR. Let $\hat{I}_t$ be the reconstructed frame produced at epoch t using the blocks available at high quality (i.e., those reused from cache or fetched/upgraded within the epoch under bandwidth and cache budgets), and let I_t be the reference frame. We define

$$q_t = \mathrm{PSNR}(\hat{I}_t, I_t) = 10 \log_{10}\left(\frac{1}{\mathrm{MSE}(\hat{I}_t, I_t) + \epsilon}\right) \tag{6}$$

where pixels are normalized to $[0, 1]$ and ϵ avoids numerical issues.

Fig. 1(d) plots the per-epoch quality trajectory q_t for a naive baseline and an H-step lookahead ideal upper bound. The naive baseline (LRU caching with greedy fetch/upgrade) exhibits repeated sharp PSNR collapses followed by slower recoveries, whereas the ideal policy largely avoids deep drops and improves the lower tail. This gap reflects how churn interacts with *coupled* resource constraints: when gaze revisits a recently evicted region, cache capacity has already forced the loss of enhancement blocks, and the per-epoch bandwidth budget limits how quickly those blocks can be re-fetched to recover quality. As a result, the system repeatedly spends budget on *re-acquiring* recently discarded content, yielding oscillatory quality and prolonged degradation around revisits.

Consequently, realizing the long-horizon caching headroom requires an online controller that is simultaneously constraint-aware and stability-aware: it must jointly decide what to keep and what to fetch under

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bandwidth and cache limits, while explicitly suppressing thrashing-induced oscillations.These observations motivate our Lyapunov-based design in §6.2, which introduces explicit stability terms and virtual-queue penalties to balance quality, switching, and resource debts.

5 Offline Asset Preparation

5.1 Offline Semantic–Saliency Co-distillation as a Leaf-aligned Sidecar

A converged 3DGS scene $\mathcal{G} = \{g_i\}_{i=1}^N$ is a high-fidelity radiance field, yet it does not expose the two quantities required by object-centric control: *what* a primitive belongs to (semantic affinity), and *how much* it matters (saliency/importance). The Deep Baker injects these two signals by learning auxiliary variables on top of a *frozen* base 3DGS model. Concretely, we freeze all geometry and appearance parameters (position, covariance/scale, rotation, opacity, SH coefficients), and optimize only (i) a per-Gaussian semantic feature $\mathbf{f}_i$ (used later to define object-level grouping and separation) and (ii) a per-Gaussian saliency/SoD scalar $s_i \in [0,1]$ (used later as a value prior in scheduling). The learned variables are stored as a leaf-aligned sidecar indexed by a canonical Gaussian index gid $= i$, leaving geometry-only hierarchy construction and node-atomic streaming intact.

For each training view v with RGB image I_v, we obtain pixel-domain supervision from off-the-shelf 2D teachers. We apply SAM3 [3], following the promptable segmentation setup established by SAM and SAM 2 [13, 25], to produce instance masks $\mathcal{M}_v = \{M_{v,k}\}$ and optionally extract region descriptors as semantic targets $\mathbf{t}_{v,k}$ (e.g., open-vocabulary embeddings). We also run a 2D saliency/attention estimator to obtain a dense map $A_v(p) \in [0,1]$. Because alpha compositing entangles multiple Gaussians along each ray, these observations are not uniquely attributable to a single primitive; we therefore supervise semantics in the rendered image domain and backpropagate to per-Gaussian variables through the differentiable 3DGS rasterizer.

To keep semantics compact and reusable, we factorize per-Gaussian features with a shared codebook. We learn a semantic codebook $S \in \mathbb{R}^{L\times D}$ and, for each Gaussian g_i, a sparse coefficient vector $\beta_i \in \mathbb{R}^L$ with only top-K non-zeros (stored by indices and weights):

$$\mathbf{f}_i = \beta_i^\top S. \tag{7}$$

Our semantic objectives are cosine-based (Eq. (17)), hence only the direction of $\mathbf{f}_i$ is relevant. We constrain each codeword to unit norm $\|S_k\|_2 = 1$ and define a normalized per-Gaussian direction

$$\hat{\mathbf{f}}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|_2 + \varepsilon}, \tag{8}$$

which is computed once per Gaussian (not per pixel). Importance is represented explicitly by s_i, avoiding ambiguity in feature magnitudes.

To make the optimization scope in Fig. 2 explicit, we partition parameters into a read-only base branch and an updatable side branch:

$$\Theta_{\text{base}} = \{\mu_i, \Sigma_i, \alpha_i, \mathbf{c}_i^{\text{SH}}\}_{i=1}^N, \qquad \Theta_{\text{side}} = \{S, \beta_i, s_i\}_{i=1}^N. \tag{9}$$

The offline co-distillation objective only updates Θ_{side}:

$$\nabla_{\Theta_{\text{base}}}\mathcal{L}_{\text{offline}} = \mathbf{0}, \qquad \Theta_{\text{side}} \leftarrow \Theta_{\text{side}} - \eta\nabla_{\Theta_{\text{side}}}\mathcal{L}_{\text{offline}}. \tag{10}$$

This exactly matches the figure-level update path: the frozen 3DGS branch is only rendered and read, while distillation gradients flow to semantic attributes and the codebook sidecar.

Given view v, the 3DGS renderer selects, for each pixel p, an ordered list of contributing Gaussians $G_p = (g_1^p, \dots, g_{K_p}^p)$ sorted by depth along the viewing ray. Let $\alpha_i^v(p)$ be the per-Gaussian opacity contribution at (v,p). Front-to-back compositing induces a visibility-gated contribution weight

$$\omega_i^v(p) = \alpha_i^v(p)\prod_{j<i}\big(1-\alpha_j^v(p)\big), \tag{11}$$

which suppresses occluded Gaussians ($\omega_i^v(p) \approx 0$). We render the semantic field as a visibility-weighted average

$$\mathbf{R}_v(p) = \frac{\sum_{i=1}^{K_p}\omega_i^v(p)\,\hat{\mathbf{f}}_{g_i^p}}{\sum_{i=1}^{K_p}\omega_i^v(p)+\varepsilon}, \tag{12}$$

and, in parallel, render the saliency/SoD field as

$$\hat{A}_v(p) = \frac{\sum_{i=1}^{K_p}\omega_i^v(p)\,s_{g_i^p}}{\sum_{i=1}^{K_p}\omega_i^v(p)+\varepsilon}. \tag{13}$$

Since $\omega_i^v(p)$ is fully determined by frozen geometry/opacity, optimizing auxiliary variables cannot alter transparency distribution; it only changes rendered semantics and saliency.

SAM masks may overlap across scales (large masks cover wholes, small masks cover parts), which introduces ambiguous consistency labels. Given an area threshold τ, we keep all fine masks ($A(M_{v,k}) \le \tau$). For coarse masks ($A(M_{v,k}) > \tau$), we enforce a small-over-large priority by sorting them in ascending area: $A(M_{v,\pi_1}) \le \cdots \le A(M_{v,\pi_C})$. Define a per-pixel occlusion indicator for the c-th coarse mask:

$$O_{\pi_c}(p) = \prod_{c'<c}\big(1 - M_{v,\pi_{c'}}(p)\big), \tag{14}$$

and activate the coarse mask only on pixels not already claimed by smaller coarse masks:

$$B_{\pi_c}(\tau,p) = M_{v,\pi_c}(p)\cdot O_{\pi_c}(p). \tag{15}$$

For fine masks, we set $B_k(\tau,p) = M_{v,k}(p)$ when $A(M_{v,k}) \le \tau$.

We optimize instance-induced semantic consistency together with saliency consistency. For two sampled pixels (p_1,p_2) in the same view, we define a binary mask-consistency label

$$y(\tau,p_1,p_2) = \mathbb{I}\left(\sum_k B_k(\tau,p_1)\,B_k(\tau,p_2) > 0\right), \tag{16}$$

and measure feature consistency by cosine similarity of rendered semantic features:

$$z(\tau,p_1,p_2) = \frac{\mathbf{R}_{v,\tau}(p_1)^\top\mathbf{R}_{v,\tau}(p_2)}{\|\mathbf{R}_{v,\tau}(p_1)\|_2\,\|\mathbf{R}_{v,\tau}(p_2)\|_2}. \tag{17}$$

We align them using a logistic distillation loss

$$\mathcal{L}_{\text{dist}}(\tau,p_1,p_2) = -y\log\sigma(z) - (1-y)\log\big(1-\sigma(z)\big), \tag{18}$$

which drives pixels belonging to the same active instance region to share similar rendered semantic directions.

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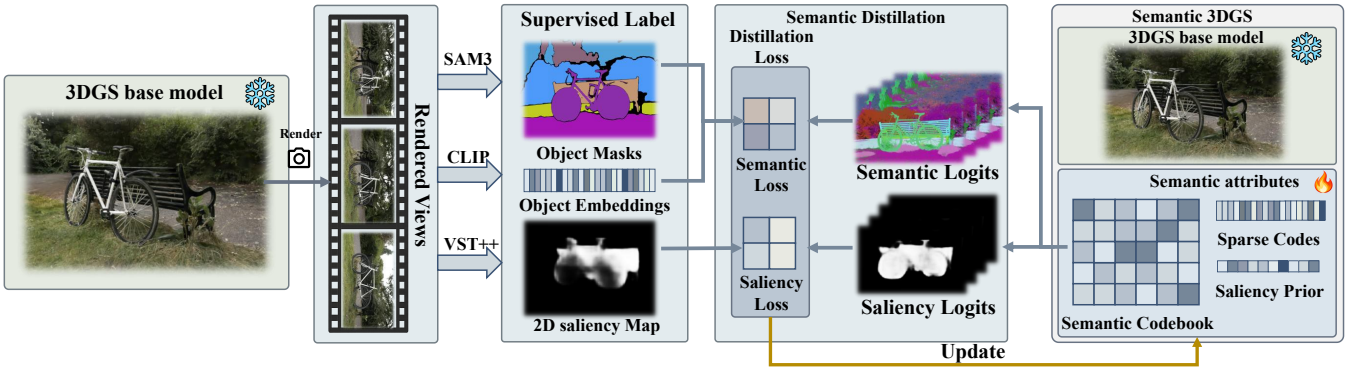

Figure 2: Offline semantic–saliency co-distillation with a frozen 3DGS base. Starting from a frozen 3DGS model (geometry/appearance unchanged), we render training views and obtain 2D supervision from off-the-shelf teachers following the promptable segmentation paradigm of SAM and SAM 2 [13, 25]: SAM3 instance masks, optional open-vocabulary region descriptors (e.g., CLIP-style embeddings), and a dense saliency/attention map. We distill two auxiliary per-Gaussian fields—semantic directions and saliency/SoD scalars—by supervising their *rendered* maps through the differentiable 3DGS rasterizer. Only a lightweight leaf-aligned sidecar is optimized and exported, keyed by the canonical Gaussian index. The sidecar contains a scene-shared semantic codebook, per-Gaussian sparse codes, and saliency priors.

For saliency/SoD, we match $\hat{A}_v(p)$ to the teacher observation $A_v(p)$ using a confidence weight $m_v(p)$ that filters low-visibility pixels and down-weights a thin boundary band:

$$m_v(p) = \mathbb{I}\left(\sum_{i=1}^{K_p} \omega_i^v(p) \geq \tau_{\text{vis}}\right) \cdot \Big(1 - \mathbb{I}(p \in \mathcal{B}_v)\Big), \tag{19}$$

where τ_{vis} is a visibility threshold and $\mathcal{B}_v$ is a boundary band derived from teacher masks (e.g., pixels within r pixels to mask edges via dilation/erosion). We denote the saliency distillation objective by

$$\mathcal{L}_{\text{sal}} = \sum_{p \in \Omega(v)} m_v(p) \left\| \hat{A}_v(p) - A_v(p) \right\|_1, \tag{20}$$

where $\Omega(v)$ denotes sampled pixels in view v.

We include an alignment regularizer to avoid degenerate solutions and to encourage ray-level semantic agreement:

$$\mathcal{L}_{\text{align}}(v) = \sum_{p \in \Omega(v)} -\log\big(\|\mathbf{R}_v(p)\|_2 + \epsilon\big). \tag{21}$$

Since $\mathbf{R}_v(p)$ is a weighted average of unit directions, $\|\mathbf{R}_v(p)\|_2 \in [0, 1]$ measures directional agreement among Gaussians contributing to pixel p.

To keep the distilled representation compact and numerically stable, we regularize sparse-code magnitude and codeword correlation:

$$\mathcal{L}_{\text{sparse}} = \frac{1}{N} \sum_{i=1}^{N} \|\beta_i\|_1, \qquad \mathcal{L}_{\text{ortho}} = \|SS^\top - I_L\|_F^2, \tag{22}$$

where I_L is the $L \times L$ identity matrix. The complete offline objective is

$$\mathcal{L}_{\text{offline}} = \lambda_d \mathcal{L}_{\text{dist}} + \lambda_s \mathcal{L}_{\text{sal}} + \lambda_a \mathcal{L}_{\text{align}} + \lambda_{sp} \mathcal{L}_{\text{sparse}} + \lambda_o \mathcal{L}_{\text{ortho}}. \tag{23}$$

In Fig. 2, the "semantic logits" and "saliency logits" panels correspond to rendered side-branch outputs derived from $\mathbf{R}_v(\cdot)$ and $\hat{A}_v(\cdot)$, visualized before packaging.

To suppress local outliers from occlusions or detector noise, we optionally smooth semantics in 3D. We define $KNN(i)$ as the K nearest neighbors of μ_i under Euclidean distance. To adapt to non-uniform Gaussian density, we use an adaptive bandwidth $\theta_i = c \cdot \frac{1}{K} \sum_{j \in KNN(i)} \|\mu_i - \mu_j\|_2$, and update

$$\mathbf{f}_i \leftarrow \frac{\sum_{j \in KNN(i)} \exp\left(-\frac{\|\mu_i - \mu_j\|_2^2}{2\theta_i^2}\right) \mathbf{f}_j}{\sum_{j \in KNN(i)} \exp\left(-\frac{\|\mu_i - \mu_j\|_2^2}{2\theta_i^2}\right)}. \tag{24}$$

The same kernel can be applied to $\{s_i\}$ if desired.

To bridge soft optimization and compact export, we adopt a quantization-aware strategy. During training, we use a differentiable approximation to sparse assignment (e.g., straight-through top-K or Gumbel-Softmax with temperature annealing) so gradients flow through index selection; at export time, we store hard top-K indices and quantized weights for compact lookup, while the codebook S is stored once per scene. After optimization, the sidecar contains (i) saliency/SoD scalars $\{s_i\}$, (ii) hard top-K indices and weights encoding $\{\beta_i\}$, and (iii) the shared codebook S, all aligned by gid.

5.2 Hybrid Decoupling and Primitive Classification

Sec. 5.1 provides two distilled signals for each Gaussian primitive g_i: a semantic direction $\hat{\mathbf{f}}_i$ and a saliency/SoD prior $s_i \in [0, 1]$. This subsection converts these soft signals into a physical decoupling that supports hybrid-granularity streaming. Intuitively, we only "promote" a subset of primitives into object-addressable units, while leaving the rest in a stable macro representation. The output of this step is three primitive classes (Queue A/B/C) used later for packaging: macro background, ambiguous transition, and micro objects.

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We begin by measuring whether a primitive is semantically reliable in its local 3D neighborhood. Let $\mathcal{N}(i)$ be the K nearest neighbors of g_i in Euclidean space. We define a local semantic agreement score

$$r_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \max\big(0, \hat{\mathbf{f}}_i^\top \hat{\mathbf{f}}_j\big), \tag{25}$$

which suppresses isolated outliers that accidentally obtain high saliency (e.g., thin occluders or detector noise).

To form micro objects, we cluster only the primitives that are both salient and locally reliable, using a joint semantic–geometric distance. For any two candidate primitives i and j, we define

$$d(i,j) = \underbrace{\frac{\|\mu_i - \mu_j\|_2^2}{\rho^2}}_{\text{geometric proximity}} + \lambda \underbrace{\big(1 - \hat{\mathbf{f}}_i^\top \hat{\mathbf{f}}_j\big)}_{\text{semantic mismatch}}, \tag{26}$$

where μ_i is the Gaussian center, ρ sets the spatial scale, and λ balances geometry versus semantics. We run a density-based clustering (e.g., DBSCAN/HDBSCAN) under $d(\cdot,\cdot)$ and obtain a set of connected components, each representing a candidate object.

Primitive classification into Queue A/B/C (no equations). Using the distilled saliency prior s_i, the local semantic reliability r_i, and the clustering results as object hypotheses, we partition primitives into three queues with distinct roles in the hybrid representation.

Queue A (Macro background). A primitive is assigned to Queue A if it does not belong to any valid object cluster, typically because it is low-saliency, semantically diffuse, or forms only tiny fragments. Queue A serves as the environmental substrate and will remain in the macro octree representation, supporting stable progressive LoD streaming.

Queue B (Transition / ambiguous band). Some primitives exhibit high saliency but fail the coherence or clustering criteria (e.g., along object boundaries, mixed materials, or teacher uncertainty). We keep these primitives in Queue B as a conservative transition band: they are not promoted to independent micro objects (to avoid fragmentation), yet they are not treated as ordinary background either. In the next subsection, Queue B is stored with the macro stream but can be scheduled with higher priority or refined progressively.

Queue C (Micro objects). A primitive is assigned to Queue C if it belongs to a sufficiently large and semantically coherent cluster produced by the joint semantic–geometric grouping above, and the cluster carries non-trivial total saliency. Queue C therefore contains only primitives that are (i) consistently grouped by semantics, (ii) spatially compact, and (iii) collectively important under attention. These primitives will be promoted into object-addressable micro bundles.

For each micro object cluster (Queue C), we summarize object-level descriptors that will be consumed by the online predictor and controller. We compute a semantic prototype and an object saliency prior by aggregating primitive-level fields:

$$\bar{\mathbf{f}}_m = \frac{\sum_{i \in \mathcal{O}_m} s_i \hat{\mathbf{f}}_i}{\left\|\sum_{i \in \mathcal{O}_m} s_i \hat{\mathbf{f}}_i\right\|_2 + \varepsilon}, \qquad \bar{s}_m = \frac{1}{|\mathcal{O}_m|} \sum_{i \in \mathcal{O}_m} s_i, \tag{27}$$

together with a tight 3D bounding box from $\{\mu_i\}_{i \in \mathcal{O}_m}$ for addressing and culling. These object descriptors are the bridge between the distilled "understanding" 5.1 and the later object-centric prediction and neuro-symbolic scheduling (Sec. VI).

5.3 Atomic Unit Packaging: Macro Octree and Micro Bundles

After the decoupling step (Sec. 5.2), the scene primitives are partitioned into three queues with distinct roles. We package them into a hybrid-granularity representation that directly matches online access patterns: micro objects are fetched and upgraded at object granularity, while the background is streamed progressively at spatial LoD.

Macro background (Queue A) as an octree with quality-preserving LoD. Primitives in Queue A are organized into a geometry-driven octree, which also defines a multi-level LoD hierarchy for progressive streaming and stable coverage. Rather than naively subsampling Gaussians (which often breaks volumetric coverage and causes noticeable artifacts), we build coarser levels by *opacity-weighted clustering* within each node: primitives are grouped in 3D using a KD-tree with a level-dependent radius, and each cluster C is replaced by a representative Gaussian whose attributes are aggregated as

$$\mu^* = \frac{\sum_{i \in C} \alpha_i \mu_i}{\sum_{i \in C} \alpha_i}, \quad \mathbf{c}^* = \frac{\sum_{i \in C} \alpha_i \mathbf{c}_i}{\sum_{i \in C} \alpha_i}, \quad \alpha^* = \frac{\sum_{i \in C} \alpha_i^2}{\sum_{i \in C} \alpha_i}, \tag{28}$$

with rotation q^* computed by weighted quaternion averaging and scale chosen to cover the entire cluster,

$$\mathbf{s}^* = \frac{1}{2}\left(\max_{i \in C}(\mu_i + \mathbf{s}_i) - \min_{i \in C}(\mu_i - \mathbf{s}_i)\right), \tag{29}$$

ensuring complete spatial coverage without gaps. The resulting octree nodes serve as macro atomic units, enabling coarse-to-fine refinement under bandwidth and rendering constraints.

Transition band (Queue B). Primitives in Queue B are stored with the macro stream as a conservative boundary band; they preserve visual continuity at object boundaries while avoiding over-fragmentation. Online scheduling may prioritize Queue B higher than ordinary background when needed.

Micro objects (Queue C) as bundles. Each micro object is stored as an independent bundle, consisting of the primitives in the cluster and the object-level descriptors summarized in Eq. (27) (prototype embedding, saliency prior, and a tight 3D bounding box). This design makes objects directly addressable for intention-driven prefetching and out-of-frustum recall, without requiring any spatial traversal over background nodes.

6 Online System

6.1 Object-Grounded Probabilistic Prediction

At each time t, the online system produces an object-level distribution

$$p_{\text{total}}(o \mid t), \qquad o \in \mathcal{O}, \tag{30}$$

where $\mathcal{O}$ denotes the set of *micro objects* distilled and packaged offline (Sec. V), plus an optional background/null token. This distribution is the sole interface exported by the predictor to the controller: it ranks which objects are most worth protecting (prefetching/upgrading) *now*, under both (i) short-horizon, geometrically

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verifiable visibility and (ii) out-of-frustum recall driven by learned object-transition regularities.

Runtime state. Each frame provides the current head/eye state $s_t = (\mathbf{p}_t, \mathbf{q}_t)$ (or an equivalent gaze-point parameterization) and a history window

$$\mathcal{H}_t = \{s_{t-w}, \dots, s_t\}. \tag{31}$$

Offline assets are read-only at runtime: each object o has a compact proxy for geometric tests (e.g., a tight 3D bounding box), an object-level semantic descriptor (Eq. (27)), and a manifest that maps (o, level/chunk) to fetchable units. The predictor may additionally consume a short sequence of recent hit objects when available (defined below).

Short-horizon head/gaze forecasting in immersive media has been studied for panoramic and 360° content [38, 39]; our predictor differs in that it converts this signal into an object-level demand distribution rather than a purely view-space forecast.

Short-horizon kinematic forecast (Physics stream). We forecast a short horizon $t + \Delta$ from $\mathcal{H}_t$ using a residual formulation:

$$\hat{s}_{t+\Delta} = \hat{s}^{\text{base}}_{t+\Delta} + \Delta\hat{s}_{t+\Delta}, \tag{32}$$

where $\hat{s}^{\text{base}}_{t+\Delta}$ is a simple kinematic baseline (e.g., constant-velocity extrapolation) and $\Delta\hat{s}_{t+\Delta}$ is predicted by a motion Transformer. The model also outputs heteroscedastic uncertainty (per-dimension log-variance) which we aggregate into a scalar confidence signal:

$$\log \sigma^2_{t+\Delta} = U(\mathcal{H}_t), \qquad u_t = \text{Agg}\big(\log \sigma^2_{t+\Delta}\big). \tag{33}$$

Intuitively, u_t is small when motion is stable and the short-horizon forecast is trustworthy, and becomes large under rapid saccades, abrupt turns, or noisy tracking.

Geometric grounding to objects (Visual stream). We convert the continuous forecast $\hat{s}_{t+\Delta}$ into a discrete object hypothesis by scoring how well each object agrees with the predicted viewing geometry. Let $\hat{\mathbf{p}}_{t+\Delta}$ be the predicted eye position and $\hat{\mathbf{d}}_{t+\Delta}$ be the predicted gaze/view direction. We define the predicted ray

$$\mathbf{r}_{t+\Delta}(\tau) = \hat{\mathbf{p}}_{t+\Delta} + \tau\, \hat{\mathbf{d}}_{t+\Delta}, \qquad \tau \geq 0. \tag{34}$$

For each object o, let P_o denote its proxy (e.g., bounding box or a small set of representative Gaussians). We compute a geometric score by measuring the perpendicular distance from the ray to primitives associated with o (or to the proxy surface) and aggregating the best matches:

$$S_{\text{geom}}(o \mid t) = \text{TopMAvg}_{i \in P_o} \exp\left(-\frac{d_\perp(i; t+\Delta)^2}{2\sigma_r^2}\right), \tag{35}$$

where $d_\perp(\cdot)$ denotes the perpendicular distance to the ray and σ_r controls the spatial tolerance. To emphasize *verifiable* visibility, we optionally restrict scoring to objects whose proxy intersects the predicted frustum; objects outside receive a very low score. We then obtain an in-frustum distribution by softmax normalization:

$$p_{\text{geom}}(o \mid t) = \text{softmax}\big(S_{\text{geom}}(o \mid t)\big). \tag{36}$$

Fixation-token intention prior (Intention stream). Geometric grounding is necessarily short-term: it is tied to current predicted visibility. To support out-of-frustum recall, we build a symbolic object sequence from recent interactions. When hit-testing is available, each frame yields a discrete hit object $h_t \in \mathcal{O} \cup \{\emptyset\}$ by intersecting the current gaze ray with object proxies. We convert the raw hit stream into fixation tokens by (i) run-length merging consecutive identical hits, (ii) filtering by a minimum dwell time, and (iii) removing $\emptyset$ and other invalid/background tokens. This produces a token sequence $(o_1, \dots, o_k)$, which is fed to a causal Transformer to predict the next-object distribution:

$$p_{\text{intent}}(o \mid t) = p(o_{k+1} = o \mid o_{\leq k}) = \text{softmax}(W\mathbf{h}_k). \tag{37}$$

Because this model learns transitions in *object space* rather than instantaneous visibility, it can allocate meaningful probability mass to objects that are currently outside the frustum.

Uncertainty-gated fusion. We fuse the two distributions using the kinematic uncertainty as a gating signal. We map u_t to a mixing coefficient

$$\alpha_t = \sigma(\beta(\tau - u_t)), \tag{38}$$

and combine the two sources in log-probability space:

$$g_o(t) \triangleq \log\big(p_{\text{geom}}(o \mid t) + \varepsilon\big), \tag{39}$$

$$i_o(t) \triangleq \log\big(p_{\text{intent}}(o \mid t) + \varepsilon\big), \tag{40}$$

$$p_{\text{total}}(o \mid t) = \text{softmax}\big(\alpha_t g_o(t) + (1-\alpha_t) i_o(t)\big). \tag{41}$$

where ε avoids numerical issues. This design yields a principled division of labor: when motion is stable (u_t small), α_t is large and the predictor trusts geometric evidence; under rapid or uncertain motion (u_t large), α_t decreases and the predictor relies more on the intention prior to avoid brittle frustum-only decisions. The resulting $p_{\text{total}}(o \mid t)$ is the object-grounded probabilistic signal consumed by the Lyapunov controller in Sec. 6.2.

6.2 Uncertainty-Gated QoE Control via Lyapunov Scheduling

We schedule at discrete epochs (every Δ seconds). At epoch t, the predictor in Sec. VI.A outputs an object distribution $p_{\text{total}}(o \mid t)$ that covers both in-frustum geometric grounding and out-of-frustum recall. The scheduler then decides which data chunks to fetch next.

QoE model (quality minus decoding latency). Following the standard volumetric streaming practice, we model QoE as a trade-off between reconstructed quality and client-side decoding latency:

$$\text{QoE}(t) = f_t(x_t) - \lambda\, g_t(x_t), \tag{42}$$

where x_t is the fetch decision at epoch t, $f_t(\cdot)$ evaluates visual quality, and $g_t(\cdot)$ evaluates decoding latency (we do not include network propagation latency in the objective).

What are f_t and g_t in our system? Each chunk k belongs to an object $o(k)$ (micro bundle) or a macro node (octree LoD chunk). Offline, we precompute a rate–distortion (RD) profile for each chunk: its compressed size size_k, its predicted quality gain $\Delta Q_k(\cdot)$ under representative view states, and an estimated decode/processing time dec_k on the client. Online, we convert the object probability p_{total} into an *expected* quality gain:

$$f_t(x_t) = Q^{\text{base}}(t) + \sum_k x_k(t)\, p_{\text{total}}\big(o(k) \mid t\big)\, \Delta Q_k(t), \tag{43}$$

$$g_t(x_t) = \sum_k x_k(t)\, \text{dec}_k. \tag{44}$$

Here $x_k(t) \in \{0, 1\}$ indicates whether chunk k is fetched at epoch t, and $Q^{\text{base}}(t)$ is the quality of the currently renderable frontier (already cached).

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Resource constraints (bandwidth and client budget). Let $B(t)$ be the available bandwidth at epoch t and $C_{\max}(t)$ be the client budget (e.g., max allowable decode time within the next epoch). The feasible decision set is constrained by

$$\sum_k \text{size}_k \, x_k(t) \;\le\; B(t)\Delta, \tag{45}$$

$$\sum_k \text{dec}_k \, x_k(t) \;\le\; C_{\max}(t). \tag{46}$$

Lyapunov scheduling (drift-plus-penalty). We maintain two virtual queues to track resource deficit over time:

$$Z_B(t{+}1) = \Big[Z_B(t) + \sum_k \text{size}_k x_k(t) - B(t)\Delta\Big]^+, \tag{47}$$

$$Z_C(t{+}1) = \Big[Z_C(t) + \sum_k \text{dec}_k x_k(t) - C_{\max}(t)\Big]^+. \tag{48}$$

Define the Lyapunov function $L(t) = \frac{1}{2}\big(Z_B^2(t) + Z_C^2(t)\big)$. At each epoch, we choose x_t by maximizing a drift-plus-penalty surrogate:

$$x_t \in \arg\max_{x \in \mathcal{X}(t)} J_t(x), \tag{49}$$

$$J_t(x) \triangleq V(f_t(x) - \lambda g_t(x)) - Z_B(t)\, A_B(x) - Z_C(t)\, A_C(x), \tag{50}$$

$$A_B(x) \triangleq \sum_k \text{size}_k x_k, \qquad A_C(x) \triangleq \sum_k \text{dec}_k x_k. \tag{51}$$

where V controls the quality–stability trade-off.

Output of the scheduler (what is produced each epoch). Eqs. (49)–(50) output a binary decision vector x_t. We convert it into an explicit request set

$$\mathcal{Q}(t) = \{\, k \mid x_k(t) = 1 \;\wedge\; k \notin \mathcal{M}(t) \,\}, \tag{52}$$

where $\mathcal{M}(t)$ is the client cache index. Each $k \in \mathcal{Q}(t)$ corresponds to a uniquely identifiable asset (either a micro-object bundle chunk or a macro octree-node LoD chunk) and is therefore directly actionable by the runtime.

Computational complexity. Let N be the number of candidate chunks considered at epoch t (after restricting to the active set driven by $p_{\text{total}}(\cdot \mid t)$ and the currently relevant macro nodes). The scalar terms in $J_t(x)$ depend only on cached per-chunk constants $(\text{size}_k, \text{dec}_k, \Delta Q_k)$ and the current queue lengths $(Z_B(t), Z_C(t))$. Thus, computing a per-chunk priority score requires $O(1)$ time, and scoring all candidates costs $O(N)$ time.

To obtain a feasible x_t under the constraints in Eqs. (45)–(46) (and any precedence rules encoded by $\mathcal{X}(t)$), we use a greedy packing procedure: we rank candidates by a priority (or priority-to-cost) score derived from $J_t(\cdot)$ and scan the ranked list until the epoch budgets are exhausted. The sorting dominates the runtime, yielding

$$T_{\text{epoch}} = O(N \log N), \qquad M_{\text{epoch}} = O(N). \tag{53}$$

Finally, constructing the request set $\mathcal{Q}(t)$ in Eq. (52) is a single pass over candidates with $O(N)$ time, assuming $\mathcal{M}(t)$ supports $O(1)$ membership queries. Operating at semantic-object / node granularity keeps N small compared to per-primitive decisions, enabling fine-grained bandwidth-efficient scheduling without incurring prohibitive online overhead.

7 Evaluation

7.1 Implementation Details

We implement OC-GS in PyTorch with differentiable Gaussian rasterization [12]. The offline semantic–saliency distillation stage is performed on a workstation equipped with an Intel Core i9-13900K CPU and a single **NVIDIA RTX A6000 GPU (48GB VRAM)**. Starting from a frozen 3DGS scene, we distill only the lightweight side information used by OC-GS and then pack the result into our object-centric mixed-granularity asset, while keeping the base Gaussian geometry and appearance unchanged.

For online experiments, the desktop workstation acts as the streaming server and runs both OC-GS and the reproduced baseline schedulers, while a **Meta Quest** headset serves as the client for interactive playback and transmission measurements. Unless otherwise stated, all streamed methods share the same server–client transmission loop, synchronized frame clocks, empty-cache initialization, warm-up stage, and fixed-length evaluation window. All reported metrics are averaged over three independent runs, and we use the same logging and replay pipeline across methods for fair comparison.

7.2 Experimental Setup and Baselines

We evaluate our method on a diverse suite of 3D scenes, including large-scale unbounded environments from the Mip-NeRF 360 dataset and cluttered indoor scenes. To simulate realistic 6DoF streaming conditions, we collected eye-tracking traces from 10 participants using a Meta Quest Pro headset. The traces capture a wide range of viewing behaviors, including rapid saccades, smooth pursuit, and fixation, with an average duration of 60 seconds per scene. The network conditions are emulated using trace-driven bandwidth logs from commercial 5G networks, with an average throughput of 50 Mbps and a standard deviation of 15 Mbps to test system robustness.

We compare OC-GS against three categories of state-of-the-art methods to validate the necessity of object-centric granularity. First, we compare against **Vanilla 3DGS**[12], which requires the entire scene to be locally available; this serves as the theoretical upper bound for rendering quality (PSNR) but fails in startup latency. Second, we compare against spatial-adaptive approaches, specifically **LapisGS** [29], L3-GS [31], and A3FR [36]. These methods utilize hierarchical spatial partitioning to stream content based on geometric proximity to the camera but lack semantic awareness. Third, we implement a **Heuristic Gaze-Streaming** baseline, which modifies Octree-GS to prioritize spatial nodes intersecting the 2D gaze point. This baseline isolates the contribution of our Neuro-Symbolic Controller by demonstrating that heuristic fetching without intention prediction and Lyapunov stability guarantees leads to buffer thrashing. Performance is measured using standard PSNR and Foveated-PSNR, which weights reconstruction error by the ground-truth object saliency to reflect perceptual fidelity on attended targets. To quantify cache efficiency, we explicitly measure the **Refetch Ratio**, which can be calculated as:

$$\eta_t = \frac{R_t}{E_t + c}, \tag{54}$$

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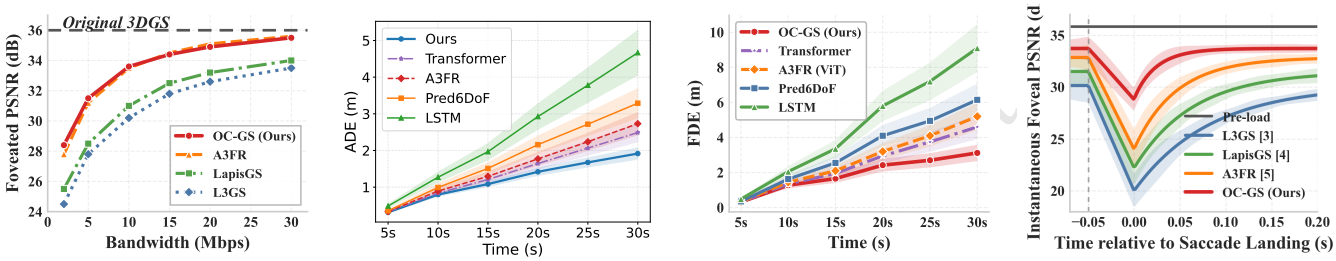
Table 1: Performance comparison with color-coded rankings (Best in red, Second in yellow). Note: Original 3DGS is excluded from ranking as it serves as the Upper Bound.

Method	Fov-PSNR ↑	Standard PSNR ↑	SSIM ↑	Refetch Ratio ↓
Original 3DGS	35.82	35.82	0.982	N/A
L3GS[31]	30.15	29.42	0.895	8.4%
LapisGS[29]	31.50	30.10	0.910	4.2%
A3FR[36]	32.85	**30.85**	0.925	5.4%
Ours	**33.72**	30.16	**0.948**	**0.8%**



where R_t denotes the number of refetch events at epoch t—i.e., chunks that have been downloaded before but must be downloaded again because they were previously evicted and are missing when requested—E_t denotes the number of evictions (cache turnovers) during the same epoch, and c is a small constant for numerical stability to avoid division by zero. In practice, we compute R_t and E_t by maintaining a per-chunk access log keyed by unique chunk IDs (macro-octree nodes and micro-object bundles), and we report the trace-averaged refetch ratio $\bar{\eta} = \frac{1}{T}\sum_{t=1}^{T} \eta_t$. Lower η indicates reduced cache thrashing and thus higher effective bandwidth utilization.

In addition to standard PSNR, we evaluate Foveated-PSNR to reflect perceptual fidelity on attended semantic targets. Let $S_t(p) \in [0, 1]$ be the ground-truth saliency weight at pixel p in frame t (derived from object saliency and normalized over the image). We define the saliency-weighted distortion

$$\mathrm{MSE}^{\mathrm{fov}} t = \frac{\sum pS_t(p), \left(\hat{I}t(p) - I_t(p)\right)^2}{\sum pS_t(p) + \epsilon}, \tag{55}$$

and compute

$$\text{Fov-PSNR}t = 10 \log 10! \left(\frac{1}{\mathrm{MSE}_t^{\mathrm{fov}} + \epsilon} \right), \tag{56}$$

where ϵ is a small constant. Unless otherwise specified, we average metrics over all frames in the session and across participants.

Table 2: Ablation Study. We evaluate the impact of each core component by incrementally enabling Semantic Granularity, Intention Prediction, and Lyapunov Stability. Saccade Latency measures the delay in regaining high fidelity after a rapid eye movement.

Method / Variant	Fov-PSNR (dB) ↑	Refetch Ratio (%) ↓	Saccade Latency (ms) ↓
(a) Baseline (Spatial-only)	30.45	6.2%	320
(b) OC-GS w/o Prediction	32.10	4.8%	285
(c) OC-GS w/o Stability	33.50	5.1%	34.2
(d) OC-GS (Full System)	**33.72**	**0.8%**	**33.5**

7.3 Ablation Study

Table 2 reports an ablation study in which components are incrementally enabled to isolate each contribution. Variant (a), the spatial-only baseline, schedules at octree-node granularity without semantic awareness, yielding the lowest Fov-PSNR and highest motion-to-photon latency. Variant (b) introduces offline micro-object granularity while retaining reactive fetching; Fov-PSNR improves substantially owing to more targeted bandwidth allocation, yet saccade latency remains elevated in the absence of proactive prefetching. Variant (c) additionally enables the intention-stream predictor for out-of-frustum recall, effectively eliminating saccade delay; however, without queue regulation the greedy prefetch policy incurs a high refetch ratio (5.1%) under coupled bandwidth and buffer constraints. The full system (d) adds the Lyapunov solver, which explicitly regularizes queue dynamics and suppresses cache thrashing, reducing the refetch ratio to 0.8% while preserving the quality and latency gains established by the preceding components.

Impact of Object-Centric Granularity. Switching from octree-node scheduling (a) to micro-object granularity (b) yields a +1.65 dB improvement in Fov-PSNR. This gain demonstrates that decoupling semantically attended targets from background geometry enables

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the limited bandwidth budget to be preferentially allocated to perceptually salient primitives, directly alleviating the granularity mismatch inherent in spatial partitioning.

Impact of Intention Prediction. Variant (b) performs fetching solely through instantaneous gaze ray-casting, which, despite improving reconstruction quality, incurs high saccade latency (285 ms) because content is only requested after fixation is established. Incorporating the intention stream in Variant (c) enables out-of-frustum recall, allowing the client to prefetch likely targets before they enter the foveal region. The resulting reduction in motion-to-photon delay demonstrates the necessity of modeling demand as a semantic, forward-looking signal rather than as a reactive spatial trigger.

Impact of Lyapunov Stability Control. Variant (c) adopts a greedy fetch policy that selects the highest-probability objects until the bandwidth budget is exhausted. While this attains competitive Fov-PSNR and near-zero saccade latency, it suffers from a high refetch ratio (5.1%), indicating pronounced cache thrashing driven by probabilistic jitter in the predicted demand distribution. Enabling Lyapunov optimization in the full system (d) introduces explicit queue-stability penalties (Eq. 46), suppressing the refetch ratio to 0.8% at the cost of only a marginal reduction in peak reconstruction quality. This improvement translates directly to markedly smoother streaming behavior under bandwidth fluctuations, validating the necessity of the control-mismatch correction identified in §2.

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A Offline Distillation Details

This appendix section will expand the offline distillation pipeline, training objectives, and object packaging details used by OC-GS.

A.1 Distillation Pipeline and Objectives

This subsection is reserved for the detailed distillation workflow, loss design, and hyperparameter settings.

A.2 Queue A/B/C Partition Rules

This subsection is reserved for the criteria that distinguish Queue A, Queue B, and Queue C, including semantic confidence, saliency, spatial scale, asset size, and their runtime roles.

A.3 Object Packaging Statistics

This subsection is reserved for package-level statistics, object-size distributions, and compression or utility analysis after object packaging.

B Gaze Prediction Details

This appendix section will provide the detailed design and diagnosis of the gaze prediction module.

B.1 Architecture and Training Setup

This subsection is reserved for the predictor architecture, data construction, and training configuration.

B.2 Prediction Diagnostics

This subsection is reserved for additional diagnostics beyond ADE/FDE, including object-level demand metrics when available.

C Supplementary Experiments

This appendix section will collect supporting experiments that complement the main paper.

C.1 Resource Sensitivity Analysis

This subsection is reserved for bandwidth, cache-budget, and scene-complexity sweeps.

C.2 Runtime and Memory Breakdown

This subsection is reserved for the breakdown of offline distillation, packaging, online prediction, scheduling, and memory overhead.

C.3 Additional Qualitative Cases

This subsection is reserved for more qualitative examples beyond the main-paper case study.

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